

Delhi Public School, Sector-4, Bokaro

Subject: system arch | Class: 10

Time Allowed: 1.5 Hours

Maximum Marks: 31

All questions are compulsory unless stated otherwise.

Name: _____ Roll No: _____ Class: 10 Section: _____

Section A

Answer all short answer questions. Each question carries 2 marks.

1. What is the primary function of the Control Unit (CU) in a CPU? **Easy** [2 Marks]
2. Explain the difference between Volatile and Non-Volatile memory, providing one example of each. **Medium** [2 Marks]
3. Define the term 'System Bus' and list its three primary component buses. **Hard** [2 Marks]

Section B

Answer all diagram-based questions. Each question carries 5 marks. Provide clean sketches and clear explanations where required.

1. Draw a basic block diagram of the Von Neumann Architecture. Label the CPU (including ALU and CU), Memory Unit, and Input/Output devices, clearly showing the direction of data flow and control signals. **Easy** [5 Marks]
2. Create a pipeline timing diagram for a 3-stage pipeline (Fetch, Decode, Execute) executing 4 consecutive instructions. Explain how this overlapping execution improves overall processor performance. **Medium** [5 Marks]
3. Sketch the pyramid diagram representing the Memory Hierarchy of a computer system. Label the levels (Registers, Cache, RAM, Secondary Storage) and use arrows to indicate the directions of increasing speed, cost per bit, and storage capacity. **Medium** [5 Marks]
4. Draw a block diagram illustrating the connection between the CPU, Cache Memory, and Main Memory (RAM). Use this diagram to explain the step-by-step process of data retrieval during a 'Cache Hit' versus a 'Cache Miss'. **Hard** [5 Marks]
5. Draw a detailed flowchart representing the Instruction Cycle. Ensure you include the Fetch, Decode, and Execute phases, and briefly explain the roles of the Program Counter (PC) and Instruction Register (IR) during this cycle. **Hard** [5 Marks]

Answer Key

1. The primary function of the Control Unit (CU) is to direct the operation of the processor. It fetches instructions from main memory, decodes them to determine what operation needs to be performed, and manages the flow of data and control signals between the CPU, memory, and peripheral devices.
2. Volatile memory requires continuous electrical power to maintain its stored data; when power is lost, data is erased (e.g., RAM). Non-volatile memory retains its stored data even when the device is powered down (e.g., ROM or Hard Disk Drives).
3. A System Bus is a communication pathway that connects the major components of a computer system (CPU, memory, and I/O). Its three primary components are: 1. Data Bus (transfers actual data), 2. Address Bus (specifies memory locations), and 3. Control Bus (transfers control signals/commands).
4. The diagram should depict three main blocks: 1. CPU containing the Arithmetic Logic Unit (ALU) and Control Unit (CU), 2. Memory Unit (RAM), and 3. Input/Output Devices. Arrows must show bidirectional data flow between CPU and Memory, unidirectional input to CPU, and unidirectional output from CPU. Control signals from the CU should point to all units.
5. The timing diagram should have Clock Cycles (1 to 6) on the horizontal axis and Instructions (I1 to I4) on the vertical axis. Cycle 1: I1 (Fetch). Cycle 2: I1 (Decode), I2 (Fetch). Cycle 3: I1 (Execute), I2 (Decode), I3 (Fetch). Cycle 4: I2 (Execute), I3 (Decode), I4 (Fetch). Cycle 5: I3 (Execute), I4 (Decode). Cycle 6: I4 (Execute). Explanation: Pipelining improves performance (throughput) by executing different phases of multiple instructions simultaneously, reducing the average clock cycles per instruction (CPI).
6. The diagram must be a pyramid. From top (smallest, fastest) to bottom (largest, slowest): 1. Registers, 2. Cache Memory, 3. Main Memory (RAM), 4. Secondary Storage (SSD/HDD). An arrow pointing upwards should be labeled 'Increasing Cost per Bit' and 'Increasing Speed/Decreasing Access Time'. An arrow pointing downwards should be labeled 'Increasing Storage Capacity'.
7. The diagram should show the CPU connected directly to Cache, and Cache connected to Main Memory. Process: 1. CPU requests data. 2. CPU checks Cache first. 3. Cache Hit: Data is found in Cache and immediately sent to CPU (fast). 4. Cache Miss: Data is not in Cache, so the system fetches it from the slower Main Memory, copies it into Cache for future use, and delivers it to the CPU.
8. The flowchart must show a continuous loop: Start -> Fetch Instruction -> Decode Instruction -> Execute Instruction -> Store Results -> loop back to Fetch. Explanation: The Program Counter (PC) holds the address of the next instruction to be fetched. The Instruction Register (IR) holds the current instruction while it is being decoded and executed.